06/12/24 M/S

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[23CSE303]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

SOFTWARE ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Differentiate between verification and validation	L2	CO1	**02**
	b.	Define Non functional requirements	L1	CO2	**02**
	c.	Why is user interface design important for software usability?	L2	CO3	**02**
	d.	What is the main objective of software testing?	L2	CO4	**02**
	e.	What is Earned Value (EV) in project management?	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	List and describe the different software process models.	L1	CO1	**05**
		OR			
	b.	Evaluate the importance of prototyping in software engineering. In what cases would it be the best choice for development?	L5	CO1	**05**
3.	**a.**	Discuss a scenario where you would implement Spiral development instead of the Waterfall model.	L3	CO1	**05**
		OR			
	b.	Describe the key differences between functional and non-functional requirements in software engineering.	L1	CO2	**05**
4.	**a.**	What is a Software Requirements Document (SRS)? Describe its main components.	L2	CO2	**05**
		OR			
	b.	How would you apply architectural design principles to design a distributed system?	L3	CO3	**05**
5.	**a.**	How would you redesign a user interface to improve accessibility for visually impaired users?	L3	CO3	**05**
		OR			
	b.	Analyze the advantages and limitations of black box testing in comparison to white box testing	L4	CO4	**05**
6.	**a.**	Explain how system testing ensures that a software application meets its requirements.	L2	CO4	**05**
		OR			
	b.	Describe the role of regression testing in maintaining software quality after changes or updates.	L2	CO4	**05**
7.	**a.**	Explain the difference between direct and indirect software measures.	L2	CO5	**05**
		OR			

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	b.	How is a Work Breakdown Structure (WBS) used in software project scheduling?	L3	CO5	05
		PART-C		**6 X 10=60**	
8.	a.	Describe the software development life cycle (SDLC) in detail. How does each phase contribute to the creation of high-quality software?	L1	CO1	10
		OR			
	b.	Critically assess the Spiral model in managing risks in software development. How does it outperform other models in terms of flexibility and iterative improvement?	L5	CO1	10
9.	a.	Explain the Software Requirements Document (SRS) in detail. Discuss the key elements that are included and how each section contributes to the project's success.	L2	CO2	10
		OR			
	b.	How would you apply the feasibility study to determine the viability of a new software project? Provide a step-by-step explanation, including technical, operational, and economic feasibility considerations.	L3	CO2	10
10.	a.	Explain the concept of modular design and discuss how it enhances the maintainability and scalability of complex systems.	L2	CO3	10
		OR			
	b.	Create a system architecture using a combination of architectural styles for an IoT-based healthcare monitoring system. Explain your design choices	L5	CO3	10
11.	a.	How would you apply equivalence partitioning to test an input field that accepts values between 1 and 100?	L3	CO4	10
		OR			
	b.	Analyze how black box testing techniques like equivalence partitioning and boundary value analysis can improve the efficiency of test case design for a login system.	L4	CO4	10
12.	a.	Evaluate the benefits and challenges of implementing white box testing in a software system with high code complexity.	L5	CO4	10
		OR			
	b.	Evaluate the significance of software measures and measurements in improving software quality.	L5	CO5	10
13.	a.	Use the COCOMO (Constructive Cost Model) to estimate the effort for a software project.	L3	CO5	10
		OR			
	b.	What are the four types of software maintenance, and how do they differ?	L2	CO5	10